

Camera Calibration and 3D Scene Reconstruction from image sequence and rotation sensor data

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Abstract

We address the problem of using external rotation information together with uncalibrated video sequences for improved calibration and 3D-reconstruction. It is shown that with a combination of a linear and a statistical approach a full calibration of the cameras can be computed from the fundamental matrices and the rotation data alone. The statistical calibration exploits some common constraints of cameras. It is analyzed which constraints are needed for the calibration of a freely moving camera. Furthermore we show that this calibration technique improves metric 3D scene reconstruction and avoids projective skew.

1 Introduction

Scene reconstruction from uncalibrated image sequences is an active research topic. One of the main tasks of the reconstruction from uncalibrated images is the calibration of the intrinsic camera parameter to reach a metric reconstruction. During the last decade we have seen a lot of progress in camera selfcalibration. Most of the approaches use image data alone and don't incorporate additional information. They suffer from degeneracies and tend to be sensitive to noise because they have to rely on the image content only.

On the other hand, in many applications we have additional information from other sensors available for example future cars will be equipped with fixed or even rotating or zooming cameras for driver assistance, where at least partial orientation and translation information is available. Another type of application is surveillance with rotating and zooming cameras. This additional data could be integrated to

improve camera calibration.

In this contribution we will discuss the possibilities to use this external orientation information for selfcalibration of freely moving cameras. We will first review the literature and existing image-based selfcalibration methods in sections 2 and 3. Selfcalibration from image and rotation data will be discussed in detail in section 4. Finally we will discuss some experiments and conclude.

2 Previous work

Camera calibration is still a current subject of research in the field of computer vision. The first major work on selfcalibration of a camera by simply observing an unknown scene was presented in [12]. It was proven that selfcalibration was theoretically and practically feasible for a camera moving through an unknown scene with constant but unknown intrinsics. Since that time various methods have been developed.

Methods for the calibration of rotating cameras with unknown and varying intrinsic parameters were developed in [6]. Camera selfcalibration from unknown general motion and constant intrinsics has been discussed in [12]. For varying intrinsics and general camera motion the selfcalibration was proven by [10, 14]. All these approaches for selfcalibration of cameras only use the images of the cameras themselves for the calibration.

Furthermore there are some interesting approaches for camera calibration with some structural constraints on the scene. Rother and Carlsson [7] proposed to jointly estimate fundamental matrices and homographies from a moving camera that observes the scene and some reference plane in the scene simultaneously. The homography induced

by the reference plane generates constraints that are similar to a rotation sensor and selfcalibration can be computed linearly. This approach needs information about the scene in contrast to our approach which applies constraints only to the imaging device.

Only a few approaches exist to combine image analysis and external rotation information for selfcalibration. There is a lot to be done in this field since there are many applications where this situation occurs such as: cameras mounted in cars for driver assistance, robotic vision heads, surveillance cameras or PTZ-cameras for video conferencing often provide rotation information. In [13, 11] the calibration of rotating cameras with constant intrinsics and known rotation was discussed. They use non-linear optimization to estimate the camera parameters. A linear approach for an arbitrarily moving camera was shown in [1, 2]. That approach is able to linearly compute a full camera calibration for a rotating camera and a partial calibration for freely moving camera.

In this paper we will address one of the few cases which have not yet been explored, that of a freely moving camera with varying intrinsics and known rotation information. We will show that orientation information is helpful for camera calibration. In extension to [1, 2] we will present an approach which is able to fully calibrate freely moving cameras by using prior knowledge.

3 Selfcalibration from image sequences

In this section we will explain some general notation and summarize previous attempts for selfcalibration.

The projection of scene points onto an image by a camera may be modeled by the equation $m = PM$. The image point in projective coordinates is $m = [x, y, w]^T$, where $M = [X, Y, Z, 1]^T$ is the world point and P is the 3×4 camera projection matrix. The matrix P is a rank-3 matrix. If it can be decomposed as $P = K[R^T | -R^T t]$ the P-matrix is called metric, where the rotation matrix R and the translation vector t represents the Euclidian transformation between the camera and the world coordinate system. The intrinsic parameters of the camera are contained in the matrix K which is an upper trian-

gular matrix

$$K = \begin{bmatrix} f & s & c_x \\ 0 & a \cdot f & c_y \\ 0 & 0 & 1 \end{bmatrix}, \quad (1)$$

where f is the focal length of the camera expressed in pixel units. The aspect ratio a of the camera is the ratio between the size of a pixel in x and the size of a pixel in y. The principal point of the camera is (c_x, c_y) and s is a skew parameter which models the angle between columns and rows of the CCD-sensor.

The camera selfcalibration problem can be split into two major cases

1. Selfcalibration of a rotating camera.
2. Selfcalibration of a freely moving camera.

3.1 Selfcalibration of rotating cameras

At first we want to summarize the calibration techniques for rotating cameras. For rotating cameras the camera transformation for camera i can be described by a 3D projective homography $H_i = K_i R_i^T$ where K_i is the calibration matrix of camera i and R_i^T is the rotation of camera i . Therefore the transformation between two rotated cameras i and j with the same optical center is given by

$$H_{j,i}^\infty = K_i R_{j,i} K_j^{-1}, \quad (2)$$

where $R_{j,i}$ denotes the rotation between camera j and camera i . The homography $H_{j,i}^\infty$ rotational compensates image i w.r.t. image j .

It can be seen that with known rotation and one known calibration K_0 of camera 0 can be (2) rewritten

$$K_i = H_{0,i}^\infty K_0 R_{0,i}^T. \quad (3)$$

to compute a calibration of all other cameras of the sequence even in the case of varying calibration.

Often all calibration matrices K_i are unknown or only some constraints on K_i 's are given. We summarize the technique from [6] where the internal and external camera calibration can be computed from images even in the case of varying internal parameters. In [6] the infinite homography constraint (IHC) [5] is used to get a set of linear equations by using some constraints like zero camera skew. Afterwards a nonlinear and maximum likelihood or a maximum a posteriori (MAP) optimization is computed.

In case of noise the approach of [6] sometimes fails to compute the calibration K_i because of numeric problems. Another disadvantage is that not all intrinsics are allowed to vary in this approach.

3.2 Selfcalibration and 3D reconstruction from general motion

For freely moving cameras and general 3D-scenes, the relation between two consecutive frames is described by the fundamental matrix [5] if the camera is translated between these frames otherwise the relation between two consecutive frames is described by the homography $H_{j,i}^\infty$ from (2). The fundamental matrix $F_{j,i}$ maps points from camera j to lines in camera i . Furthermore the fundamental matrix can be decomposed into a homography $H_{j,i}^\pi$ which maps over the plane π and an epipole e which is the projection of the camera center of camera j in camera i

$$F_{j,i} = [e]_x H_{j,i}^\pi, \quad (4)$$

where $[\cdot]_x$ is the cross product matrix. The epipole is contained in the null space of F : $F_{j,i} \cdot e = 0$.

The fundamental matrix is independent from any projective skew. This means that the projection matrices P_j and P_i lead to the same fundamental matrix $F_{j,i}$ as the projectively skewed projection matrices $\tilde{P}_j$ and $\tilde{P}_i$ [5]. This property poses a problem when computing camera projection matrices from Fundamental matrices. Most techniques for calibration of translating and rotating cameras first estimate the projective camera matrices P_i and the positions M_k of the scene points from the image data with a Structure-from-Motion approach. The estimated projection matrices P_i and the reconstructed scene points may be projectively skewed by a projective transformation $H_{4 \times 4}$. Then the skewed projection matrices $\tilde{P}_i = P_i H_{4 \times 4}$ are estimated and inversely skewed scene points $\tilde{M} = H_{4 \times 4}^{-1} M$ are also estimated. For uncalibrated cameras one cannot avoid this skew and selfcalibration for the general case is concerned mainly with estimating the projective skew matrix $H_{4 \times 4}$ via the dual image of the absolute conic [5] or the absolute quadric [10, 14].

4 Full calibration of freely moving cameras with known rotation

In this section we will develop novel techniques to use available external orientation information for

full camera selfcalibration of freely moving cameras. It is assumed that the alignment in time between orientation data and camera data is given.

4.1 Selfcalibration of a rotating camera with known rotation

The selfcalibration of a rotating camera with known orientation was discussed in [1, 2]. It exploits the given rotational information to overcome the limitations on the number of varying intrinsics and the problems caused from noise during computation in [6]. Furthermore in [1, 2] it was shown that the given orientation information linearizes the calibration problem for a rotating camera. The influence of registration and rotation inaccuracies was analyzed. It was shown that registration and rotation inaccuracies lead to noise in the estimation of some of the intrinsics namely the principal point (c_x, c_y) and the skew s of the camera. The estimation of the focal length and the aspect ratio is noise robust for an ordinary range of registration and rotation error. An approach to improve the calibration accuracy was proposed. The linear approach is first used to compute focal length f and aspect ratio a of the camera linearly, afterwards a maximum a posteriori estimation computes a full calibration robustly.

4.2 Selfcalibration from general motion

The case of general motion and external rotation information is also investigated by [1, 2]. The fact is used that the fundamental matrix as opposed to projection matrix is not affected by projective skew, therefore it can be used to calibrate the cameras. It is shown that similar to the case of a rotating camera the following equation is valid for a freely moving camera and the estimated Fundamental matrices $\tilde{F}_{j,i}$

$$\begin{aligned} 0_{3 \times 3} &= [e]_x K_i R_{j,i} - \tilde{F}_{j,i} K_j \\ &= [e]_x \tilde{K}_i R_{j,i} - F_{j,i} K_j \end{aligned} \quad (5)$$

with $\tilde{K}_i = \rho_{j,i}^{-1} K_i$, which is linear in the intrinsics of camera j and the scaled intrinsics of camera i in conjunction with the scale $\rho_{j,i}^{-1}$. Eq. (5) provides six linearly independent equations for the scale and the intrinsics of the cameras. From a counting argument it follows that the solution is never unique if no constraints for the scales $\rho_{j,i}^{-1}$ or the intrinsics of the cameras are available.

Therefore [1, 2] use common prior knowledge about the principal point of the cameras to compute a partial camera calibration.

4.3 Properties of the linear calibration of freely moving cameras

In this subsection we discuss the abilities of the linear approach in detail.

At first we inspect which sets of constraints provide a unique solution for (5). We can constrain the K_i 's by different parameters settings:

- known principal point: The solution for the focal length, aspect ratio and skew is unique if we use two image pairs.
- known skew and principal point: We can estimate the focal length and aspect ratio directly from a single fundamental matrix.
- known skew, known aspect ratio and principal point: The solution for the focal length is unique for one image pair. Note that this case is also linear in the case of unknown rotation [5].

These constraints can be applied for efficient self-calibration in case of general camera trajectory. The most common constraint of zero skew doesn't provide a unique solution for focal length, aspect ratio and principal point.

With respect to the evaluated noise robustness for rotating cameras the most successful set of constraints is known skew and principal point. In this case the aspect ratio and the focal length can be estimated linearly from an image pair (j, i) with (5). The noise robustness of the focal length and the aspect ratio is similar to the robustness for a rotating camera [1].

Furthermore the known rotation can be used to detect critical motion sequences for the solution of the selfcalibration problem. Critical motion sequences mean that it is not possible to fully determine the projective skewing homography $H_{4 \times 4}$ and therefore the camera can't be calibrated completely. The calibration with known rotation has the same critical motion sequences as the general calibration problem. Pure translation of the camera can be detected with zero rotation about all axes. In this case the reconstruction is only affine. Another critical motion is planar translation of the camera and rotation about an axis perpendicular to that plane. This critical motion can be detected by measuring the or-

thogonality between the rotation axis and the camera motion plane.

4.4 Statistical calibration of a freely moving camera with known rotation

In this subsection we will introduce a novel statistical calibration technique to reach a full calibration for a freely moving camera.

The above linear approach (5) is able to robustly estimate the focal length f and the aspect ratio a . The estimation of the principal point in conjunction with the focal length f and the aspect ratio is not possible because in this case the solution of (5) is not unique. For the most cameras the principal point is located close to the image center and the skew is normally zero for digital cameras. Therefore our novel calibration technique uses the prior knowledge about the principal point and the skew to extend the linear partial calibration for focal length f and the aspect ratio a to a full calibration by a statistical optimization.

Let us consider that the noise $n_{\tilde{m}}$ on the measured image point position $\tilde{m}$ is additive and has a Gaussian distribution with zero mean and standard deviation $\sigma_{\tilde{m}}$. The measured image point $\tilde{m}$ is thus related to the true point m by:

$$\tilde{m} = m + n_{\tilde{m}} = F(\Omega) + n_{\tilde{m}}$$

where Ω contains the model parameters of the camera model and the scene and m is the real image point. The function $F(\cdot)$ represents our model of the imaging process. We here use the linear model of a pinhole camera to model the imaging process [5]. Please note that with a nonlinear $F(\cdot)$ one is also able to model nonlinearities of the real imaging process like radial distortion. For the imaging process of the pinhole model we get

$$\tilde{m} = PM + n_{\tilde{m}} = KR[I_{3 \times 3} - C]M + n_{\tilde{m}}$$

for the measured location $\tilde{m}$ of the image point m . Then a Maximum Likelihood estimation for the intrinsics of the camera j resp. i and the rotation between camera j and i is given by:

$$\text{MLE} = \arg \min_{K_i, R_i} \sum_{i=1}^{\#cams} \sum_{k=1}^{\#pts} \epsilon_{j,i}(m_{i,k}, m_{j,k})^2, \quad (6)$$

where $\epsilon_{i,j}(\tilde{m}_{i,k}, \tilde{m}_{j,k})$ is the weighted distance between the point $\tilde{m}_{i,k}$ in image i to the epipolar line

$\tilde{l}_{\tilde{m}_{j,k}} = F_{j,i} \tilde{m}_{j,k}$ of the corresponding point $\tilde{m}_{j,k}$ in image j depending on $F_{j,i}$. This weighted distance is

$$\epsilon_{i,j}(\tilde{m}_{i,k}, \tilde{m}_{j,k}) = \frac{\tilde{m}_{i,k}^T F_{j,i} \tilde{m}_{j,k}}{\sigma_{\tilde{m}_{i,k}} \sigma_{\tilde{m}_{j,k}} \gamma}$$

with

$$\gamma = ((\tilde{l}_{\tilde{m}_{j,k}})_1^2 + (\tilde{l}_{\tilde{m}_{j,k}})_2^2)(\tilde{m}_{i,k})_3^2$$

where $\sigma_{\tilde{m}_{i,k}}$ and $\sigma_{\tilde{m}_{j,k}}$ are the variances of the Gaussian distributions of the image measurements $\tilde{m}_{i,k}$ and $\tilde{m}_{j,k}$. The optimization parameters for the calibration are contained in the linear Fundamental matrix transformation $F_{i,j}$

$$F_{i,j} = [e]_x K_i R_{j,i} K_j^{-1}.$$

If the computation is done with normalized coordinates (6) can be approximated with

$$\text{MLE}_{app} = \arg \min_{K_i, R_i} \sum_{i=1}^{\#cams} \sum_{k=1}^{\#pts} \hat{\epsilon}_{j,i}(\tilde{m}_{i,k}, \tilde{m}_{j,k})^2, \quad (7)$$

where the distance $\hat{\epsilon}_{j,i}(\tilde{m}_{i,k}, \tilde{m}_{j,k})$ is

$$\hat{\epsilon}_{j,i}(\tilde{m}_{i,k}, \tilde{m}_{j,k}) = \tilde{m}_{i,k}^T F_{j,i} \tilde{m}_{j,k}$$

which can be computed directly from the point correspondences $\tilde{m}_{i,k}$ and $\tilde{m}_{j,k}$. The approximation error of (7) with respect to (6) is very small for normalized [3] image coordinates.

If we model the expectation pp_{pri} that the principal point probability lies close to the optical center of the camera and has a Gaussian distribution whose mean is the image center, a Maximum a Posteriori estimation of the intrinsics of camera i and j and the rotation $R_{j,i}$ between the cameras is simply given by:

$$\text{MAP}_{pp} = \text{MLE}_{app} + \lambda_{pp} \sum_{i \in cameras} d_{pp}(c^i) \quad (8)$$

where d_{pp} is the weighted distance between prior principal point pp_{pri} and current estimated principal point c^i . This distance can be computed with

$$d_{pp}(c^i) = (c^i - pp_{pri})^T \begin{bmatrix} \sigma_x^2 & 0 \\ 0 & \sigma_y^2 \end{bmatrix} (c^i - pp_{pri}).$$

where λ_{pp} is the weight of the prior knowledge and σ_x^2 , σ_y^2 the distribution parameters for the components of the principal point. Furthermore we are

able to use the given sensor orientation as a prior knowledge:

$$\text{MAP}_{ori} = \text{MAP}_{pp} + \lambda_{ori} \sum_{i \in cameras} d_{ori}(r_{est}) \quad (9)$$

with the distance $d_{ori}(r_{est})$ between estimated and measured rotation. This distance can be computed with

$$d_{ori}(r_{est}) = (1 - \langle r_{j,i}, r_{est} \rangle) + |\phi_{i,j} - \phi_{est}|$$

where $r_{j,i}$ is the rotation axis of $R_{j,i}$ and $\phi_{j,i}$ is the rotation angle about $r_{j,i}$. The estimated rotation axis is r_{est} and ϕ_{est} is the estimated rotation angle about r_{est} . Now we are able to optimize the orientation information concurrently with the calibration. This can be used to improve the orientation data. The statistical optimization is started with the linearly estimated focal length and aspect ratio and the prior knowledge about principal point and skew.

4.5 Evaluation of the statistical calibration

We tested the calibration of a moving and rotating camera by using images rendered from a photorealistic image renderer. A tilting and panning camera is moving sideways in front of a VRML-scene created from realistic 3D models of buildings (see figure 1). The focal length of the camera was fixed to 415 (in pixel). We tracked features over the image sequence with the KLT-tracker [4]. From these features we estimated Fundamental matrices for different image pairs. The rotation is the given rotation of the ground truth data. We didn't use the fixed calibration of the cameras as an additional constraint during the MAP estimation process. The linear estimated focal length has a mean relative error of 4% w.r.t. the true focal length (see figure 3 (a)) the statistical calibration has an improved error of 3% w.r.t. the true focal length.



Figure 1: Images from the sequence for Fundamental matrix calibration with simulation data.

We also tested the statistical calibration technique for a real, moving, and rotating camera. The test sequence was taken by a consumer pan-tilt-zoom camera as used in video conferencing (Sony DV-31). For the first test sequence the camera only rotates and moves during the sequence (see figure 2 left for a frame of the sequence).

The camera rotation was taken from the camera control commands, which means that we used the angles which are sent to the camera. Therefore the rotation error depends on the positioning accuracy of the pan-tilt head which is in the range of below 0.5 degrees for each axis. As reference for the zoom we manually measured the focal length of the different discrete zoom steps of the camera to calculate approximate ground truth. The focal length of the camera for the first Sequence is 860 (in pixel). We compensated the zoom-dependent radial distortion beforehand. This can be done for the different zooming steps of the camera without knowledge of the correct focal length because this only depends on the discrete zoom step. The resulting relative mean error is about 5% for the linear calibration and about 4% for the statistical calibration.

Our second test sequence also used the above mentioned video conferencing camera. During the sequence the camera is panning, tilting, zooming and moving. Some frames of the sequence are shown in figure 2. The focal length in pixel varied between 860 and 930. We also compensated the radial distortion beforehand. The calibration results are shown in figure 3 (b). The resulting error is about 3% for the linear calibration and about 2% for the statistical calibration.



Figure 2: Images from the sequence for fundamental matrix calibration with varying calibration.

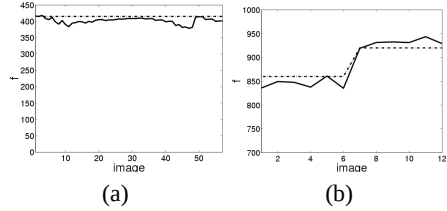


Figure 3: (a) calibration for synthetic sequence, (b): estimated focal length for varying intrinsics of real sequence. (dash-dotted: ground truth, solid: estimated values)

5 3D-Reconstruction from uncalibrated image sequences with rotation information

This section discusses the benefits of the previous full calibration of a freely moving camera for 3D reconstruction.

5.1 Metric reconstruction

The common standard reconstruction algorithm contains the following major steps:

1. Estimation of point correspondences between the images.
2. Computation of the Fundamental matrix from point correspondences between the image pairs.
3. Computation of an initial set of camera projection matrices from the estimated fundamental matrices. Please note these projection matrices are skewed projection matrices, except for some special camera configurations [5] like fixed coupled stereo cameras with fixed calibration.
4. Computation of the 3D scene points from the projection matrices and the point correspondences in the images with triangulation [5].
5. Some global bundle adjustment can be done after this computation to optimize the previous estimated cameras and scenes.
6. To reach a metric reconstruction a selfcalibration of the estimated cameras and scene has to be computed [10, 14, 5].

For image sequences with known rotation we introduce an improved reconstruction process in the following.

Using optimized Fundamental matrices After the estimation of the point correspondences from the images the statistical calibration provides a set of Fundamental matrices which optimize the epipolar geometry and the given cameras with respect to the prior knowledge used during MAP optimization. These Fundamental matrices can be used to compute the initial camera pose in step 3. The use of these Fundamental matrices give slightly better initial projection matrices. The projection matrices estimated from these Fundamental matrices are still projectively skewed. The next paragraph shows a method to overcome the projective skew of the projection matrices.

Metric projection matrices With given Fundamental matrices $F_{j,i}$ a pair of canonical projection matrices can be computed by

$$P_j = K_j[I_{3 \times 3}|0] \text{ and } P_i = K_i[[e]_x F_{j,i} + ev^T | \lambda e] \quad (10)$$

where v is any vector from $\mathbf{P}^2$ and λ can be an arbitrary scalar [5]. Using the given calibration matrices K_j and K_i from the selfcalibration we are able to compute normalized fundamental matrices with

$$\tilde{E}_{j,i} = K_i^T F_{j,i} K_j.$$

Furthermore, if we compute the singular value decomposition of the normalized Fundamental matrices $\tilde{E}_{j,i}$

$$\tilde{E}_{j,i} = U \tilde{S} V^T \text{ with } \tilde{S} = \text{diag}(a, b, 0)$$

then we compute the matrix $\hat{E}_{j,i}$ which is in Frobenius norm the closest to the Essential matrix $\tilde{E}_{j,i}$ by using diagonal matrix $S = \text{diag}\left(\frac{a+b}{2}, \frac{a+b}{2}, 0\right)$ as [5]

$$\hat{E}_{j,i} = U S V^T = [e]_x R.$$

The exact Essential matrix can be estimated using point correspondences that were normalized with the estimated calibration matrices K_j and K_i . In this case only the epipole has to be estimated. This estimation problem is very stable because only two degrees of freedom have to be determined.

From the essential matrix $\hat{E}_{j,i}$ we are able to estimate metric projection matrices. If we assume without loss of generality the projection matrix $P_j = [I_{3 \times 3}|0]$ then there are only four possible choices for the projection matrix P_i [5]. The correct pose

can be determined with oriented projective geometry. This improves the computation of the first cameras and the cameras are not projectively skewed. Therefore the estimated scene is also metric and not skewed.

This is the main goal of the rotation calibration for 3D scene reconstruction from uncalibrated image sequences, namely the full reconstruction process is metric. Therefore the space for the reconstruction process has a metric in contrast to the general approach which has to use a projective space.

5.2 Reconstruction for simulation data

We are now evaluating the calibration data from the synthetic sequence of section 4.5 (see figure 1 for sample images). The relative error of the computed camera position w.r.t. a baseline of length 1 for the first camera pair is plotted in figure 5 (left) for each axis separately. It can be seen that the error is below 6% for x and y . The larger error for z is caused from the larger uncertainty of the used reconstruction scenario. After the reconstruction process we run a stereo algorithm [8] to estimate the scene depth information from the images themselves (see figure 4 (b)). In the depth maps black means the stereo algorithm was not able to estimate any depth information for the pixel, the whiter the grayvalue is the further is the 3D point corresponding to this pixel in the scene. Due to the fact that the stereo estimation only uses two images, occlusion can't be handled around the edges of the foreground object where we have large depth estimation errors (up to 50% depth error). The depth map has a fill rate of 91%. Afterwards we measured the depth estimation error of our reconstruction. The ground truth depth map is shown in figure 4(a) and the error depth map is shown in figure 5(b). We scaled the relative error to the range of [128, 255] where black means we have no estimated value for comparison. The mean relative depth error is $4 \cdot 10^{-5}\%$ and the variance is 0.8%. Thus a very good metric reconstruction without any projective skew could be computed (see figure 6 for model).

6 Conclusions

We introduced a statistical calibration technique for a full calibration of a freely moving camera from an image sequence with known rotation of the cam-

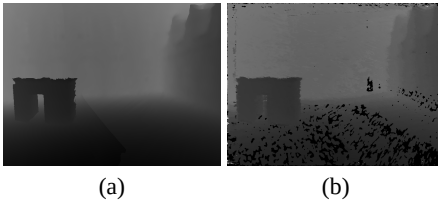


Figure 4: synthetic sequence:(a) ground truth depth map, (b) estimated depth map

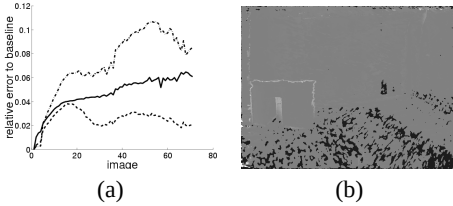


Figure 5: (a) error of estimated camera center for each component (solid x , dashed y , dash dotted z), (b) relative depth error (white $\geq 100\%$, greyvalue 128 means no depth error, black no depth estimation) Model from the sequence of a panning and tilting camera.

eras. The robustness of this calibration was measured with synthetic data with ground truth information and real data with hand measured ground truth data. It shows that the calibration is is rather stable.

Furthermore we exploit the full calibration of the freely moving camera to improve the structure-from-motion approach. These improvements are also tested with the synthetic and the real data. It shows that the estimation of the improved structure from motion process is very precise and furthermore it is a metric reconstruction.

Further the full integration of this calibration technique into the structure-from-motion approach has to be done. Another field of work is to exploit the improved rotation to stabilize the orientation sensor concurrently with the camera calibration.

References

- [1] J.-M. Frahm and Reinhard Koch, "Robust Camera Calibration from Images and Rotation Data", Proceedings of DAGM, 2003.
- [2] J.-M. Frahm and R. Koch, "Camera Calibration with known Rotation", ICCV, 2003.



Figure 6: View of the model reconstructed from the synthetic sequence

- [3] R. Hartley, "In defence of the 8-Point-Algorithm", ICCV95, Cambridge, MA, USA, 1995.
- [4] Bruce D. Lucas and Takeo Kanade, "An Iterative Image Registration Technique with an Application to Stereo Vision," *International Joint Conference on Artificial Intelligence*, pages 674-679, 1981.
- [5] R. Hartley and A. Zisserman, "Multiple View Geometry in Computer Vision" *Cambridge university press, Cambridge, 2000*
- [6] L. de Agapito et.al, "Self-calibration of a rotating camera with varying intrinsic parameters" *British Machine Vision Conference 1998*
- [7] C. Rother and S. Carlsson, "Linear multi view reconstruction and camera recovery using a reference plane", *IJCV* 49(2/3):117-141.
- [8] R.Koch et. al., "Multiviewpoint stereo from uncalibrated video sequences", *ECCV'98*, LNCS Springer, 1998
- [9] C. E. Pearson, *Handbook of Applied Mathematics*, S.898, Second Edition, Van Nostrand Reinhold Company, 1983.
- [10] B. Triggs, "Autocalibration and the Absolute Quadric", *Proceedings CVPR*, June 1997.
- [11] F. Du and M. Brady, "Self-calibration of the intrinsic parameters of cameras for active vision systems" *CVPR*, 1993.
- [12] O. D. Faugeras and M. Herbert, "The representation, recognition and locating of 3-D objects," *Intl. J. of Robotics Research*, 1992.
- [13] G. Stein, "Accurate internal camera calibration using rotation, with analysis of sources of error," *ICCV*, 1995.
- [14] Pollefeys et. al. "Selfcalibration and metric reconstruction in spite of varying and unknown internal camera parameters," *ICCV*, 1998.